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No Cap: AI Servers Are Recharging MLCC Demand // Core Research Exclusive

5 minutes

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Executive Summary

- AI server MLCC demand is inflecting sharply as GPU TDP rises with generational improvements, driving proportional growth in decoupling capacitance per rack.
- 800VDC rack power architectures shift demand toward higher-voltage, higher-ASP MLCCs, especially 100V+ parts used in 800V-to-48V conversion stages.
- Multi-die and chiplet accelerators increase MLCC content per XPU because each die and power domain requires its own decoupling

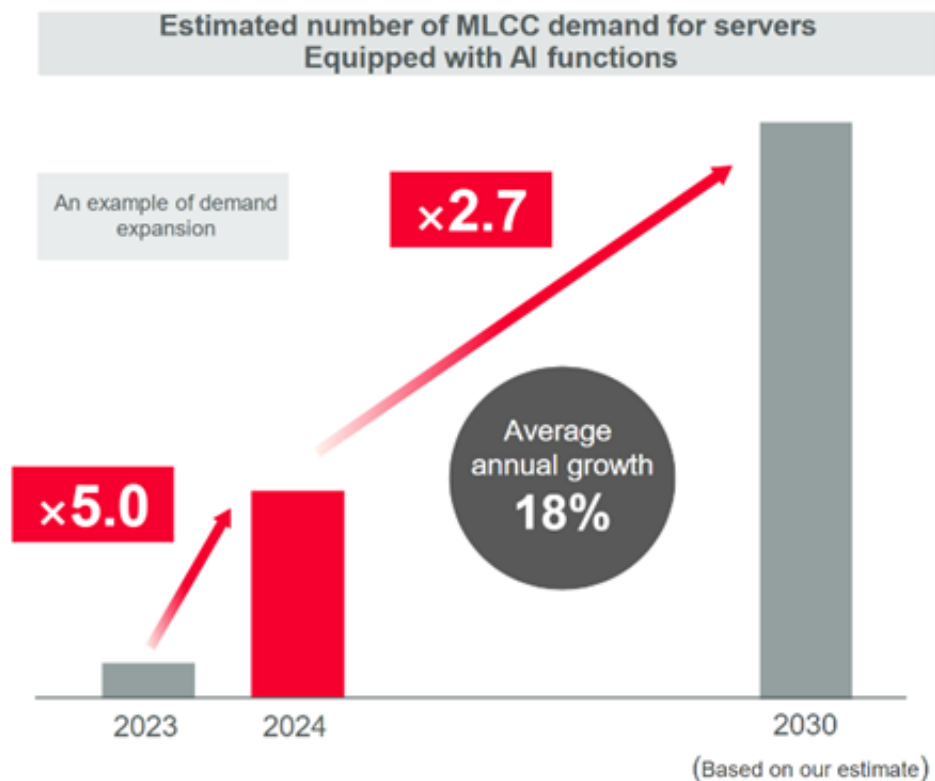
network.

- We estimate AI server MLCC demand will grow 4.8x from CY25 to CY27, with AI potentially reaching ~20% of the high-end MLCC market by 2027.
- Current MLCC demand landscape is much more structural than in 2020–2021: key vendors are not flagging over-ordering, lead times remain elevated, Murata’s capacitor book-to-bill has returned to 1.36x, and constrained new capacity supports continued tightness plus an expected ~30% high-end server MLCC price hike in 2H.

MLCC suppliers have broadly re-rated, led by SEMCO (009150.KS), up ~230% YTD, with Murata (6981.JP), TDK (6762.JP), Taiyo Yuden (6976.JP), and Yageo (2327.TW) also posting strong moves. Despite concerns around sustainability, we view AI server MLCC demand as structural and still early, with accelerating XPU shipments beginning to flow through numbers. This should support relative strength until supply catches up.

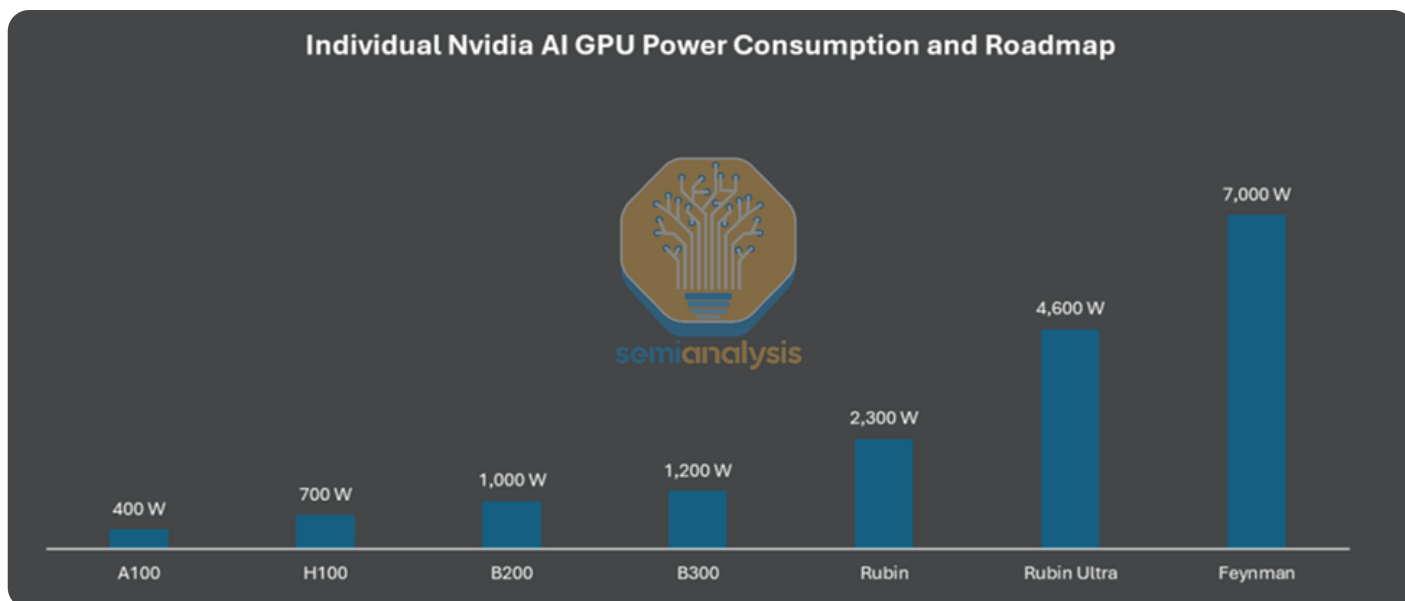
Three forces are driving MLCC content per server

An MLCC is a passive ceramic capacitor that stores and releases electrical charge. They are used throughout the server power chain, from the power supply unit all the way down to the back of the GPU die. For more details, you can refer to the Appendix at the very bottom to see where exactly MLCC sits in an AI server.



Source: Murata

- **TDP is scaling faster than Moore's Law.** GPU TDP has scaled from 400W (A100) to 700W (H100) to 1,200W (GB200) to 1,400W (GB300) to 1,800/2,300W (R200 Rubin). Each watt of TDP requires proportional decoupling capacitance. Voltage regulator MLCC content per rack scales linearly with TDP.



Source: SemiAnalysis

- **800VDC shifts MLCC demand toward higher-voltage, higher-ASP products.** Traditional 48V rack distribution used 25V-rated MLCCs. The transition to 800VDC requires 100V+ rated MLCCs inside the 800V-to-48V converter stage, which are harder to manufacture and more expensive per unit. The MLCCs inside each power conversion stage get more expensive, even if the unit count increase from the additional stage is modest. Rack-level peak shaving and load smoothing is increasingly handled by supercapacitors, not MLCCs.
- **Multi-die packaging multiplies per-XPU content.** As accelerators move to chiplet architectures, the number of power domains per socket increases, and each domain needs decoupling. For example, Vera Rubin's R200 GPU contains a compute die, C2C IO die, and NVLink IO die. Each die requires its own decoupling network. The Vera CPU adds further MLCC demand.

Source: SemiAnalysis

MLCC Demand Estimates: AI Server MLCC demand surpasses General Servers

Between CY25 and CY27 we estimate total MLCC demand for AI servers to

grow 4.8x driven by the ramp of Rubin platforms. We estimate total high end MLCCs to be around 1-1.5 trillion units shipped per year, which means by 2027 AI could represent ~20% of that market up from mid-single digits today.

Source: SemiAnalysis

For context, traditional general-purpose servers ship roughly 15M units annually at ~2,000 MLCCs each, totaling 30B units per year. NVIDIA AI server MLCC demand alone surpassed traditional server demand in CY2025 and will be 3-6x larger by CY2027.

- Our MLCC unit demand is estimated as a function of our [SemiAnalysis Accelerator Model shipments](#). Moreover, we think MLCC content per custom ASIC server is lower than per NVIDIA NVL72 rack due to lower TDP per chip and simpler rack-level power delivery.
- We estimate non-NVIDIA contributes 40-50% of additional MLCC demand on top of NVIDIA, rather than a 1:1 ratio with chip volumes.

The Last Cycle and Why Capacity Is Constrained Now

The last major MLCC upcycle came in 2020-2021, when suppliers framed the recovery as structural and broad-based across smartphones, stay-at-home electronics, 5G, industrial, and autos. By early 2021, Yageo, Samsung, and Walsin had announced 10-40% price increases, while lead times stretched from 1-3 months to roughly six months.

As consumer demand faded, EVs became the new growth story. BEVs use roughly 10,000-30,000 MLCCs versus 3,000-5,000 for ICE vehicles, pushing suppliers to invest aggressively. TDK, for example, built its Kitakami automotive MLCC factory, targeted 190% capacity growth by end-2024 versus March 2021, and raised passive component capex from ¥150B to ¥210B. But MLCC share prices began drifting lower from 2Q21, just as growth decelerated. By 2022-2024, the EV thesis weakened, consumer demand reversed, and auto customers destocked after over-ordering. Book-to-bill fell, utilization dropped, margins compressed, and the industry entered a nearly two-year trough.

Source: SemiAnalysis

Source: Bloomberg

However, we believe this cycle is different, with AI servers driving a new demand inflection. MLCC content per AI server is 8-10x higher than in traditional servers due to rising TDPs and increasingly complex power architectures. The upcoming shift to 800VDC should further improve mix: the 800V-to-48V conversion stage requires 100V+ rated MLCCs, which carry higher ASPs and are more difficult to manufacture than the 25V parts used in traditional 48V systems.

In the 2020-2021 cycle, there were warning signs of unsustainable demand drivers. On the supplier side, Murata flagged demand quality concerns in Dec 2020 earnings, telling investors "component procurements are higher than actual demand" and warning of "a phase of adjustment from March onward." TDK also warned about something similar.

In the current cycle, we reviewed earnings calls in the past 4 quarters across all four suppliers and the warning signs that existed in 2020-2021 are absent.

No supplier is flagging demand quality concerns, including the ones that did last time. Murata, which warned about over-ordering in Dec 2020, is now saying “a very big challenge is how to produce and to what extent we can meet customer demands” (April 2026 earnings). TDK, whose CEO twice flagged inventory buildup in 2020-2021, is now targeting a “tenfold increase” in AI datacenter passive component sales (April 2026 call). Semco and Yageo all called out AI driving secular demand.

Lead Time and Pricing Expectations

We examined 500 MLCC part listings across 10+ manufacturers to analyze the supply tightness narrative. From our findings Murata and Taiyo Yuden have the shortest lead times (median ~10-12 weeks), consistent with their dominant capacity positions. Samsung and Yageo/KEMET are significantly longer (median ~20 weeks), with some SEMCO server-grade parts stretching to 40+ weeks. We also see high cap server-grade lead times at 2x consumer, a level of tightness that forces even the most customer-subordinate Japanese supplier to raise prices. Evidently, Murata also announced that Capacitor BB ratios have reached back to 1.36x, the highest level since 2020.

Due to the supply tightness, and management previously hesitating to expand production lines until more recently we expect supply tightness to continue into next year given that new MLCC production lines generally take 18-24 months to come online. We expect low-end MLCC and automotive to pricing decline to moderate due to shift in production priorities to support high-end. We expect high-end server-related MLCC to see a blended ~30% price hike for the back half of this year, with Yageo and Murata being the most aggressive.

Source: SemiAnalysis

Source: SemiAnalysis, Murata

Appendix: Where MLCCs sit in the datacenter power chain

An MLCC is a passive ceramic capacitor that stores and releases electrical charge. They are used throughout the server power chain, from the power supply unit all the way down to the back of the GPU die.

- **The core problem MLCCs solve is voltage stability.** When a GPU transitions from idle to full compute, it demands current faster than the upstream power supply can respond. Without enough local capacitance, the voltage sags and the chip crashes or produces errors. High-capacitance MLCCs located near the GPU must act as current buffers for a stable system operation.

Source: Semco

- **MLCCs also sit inside the power supply itself, smoothing the voltage conversion at each stage.** A modern AI datacenter rack converts power through multiple stages: AC from the grid to high-voltage DC, down to 48V at the rack bus, to 12V or 6V at the board, and finally sub-1V at the die (more details in our [800VDC note](#)). Every conversion boundary needs MLCCs. More power, more conversion stages, more MLCCs.

Source: Semco

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